Mark schemes

Q1.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Shows integral of v with respect to t Condone omission of dt	AO1.1a	M1	$\int 0.06(2 + t - t^2)dt$
	Integrates with at least two correct terms (condone omission of constant at this stage) Any equivalent form.	AO1.1b	A1	$= 0.12t + 0.03t^2 - 0.02t^3 + c$
	Finds constant using $t = 0$ and displacement of 3. PI.	AO2.2a	M1	Using $t = 0$, $c = 3$
	Finds fully correct expression for r.	AO1.1b	A1	$r = 0.12t + 0.03t^2 - 0.02t^3 + 3$
(b)	Models problem as only force acting is that due to gravity so that $v = 0$ at highest point. (PI)	AO3.3	M1	$a = -9.8$ m s^{-2} $v = 0$ m s^{-1}
	Uses appropriate suvat formula $v = u + at$ with $u = 3.43$ and g negative or $u = -3.43$ and g positive	AO1.1a	M1	$0 = 3.43 - 9.8t_{max}$
	Finds $t_{max} = 0.35$ s (CAO) Condone addition of the 2 seconds to give $t = 2.35$ s or 2.4 s	AO1.1b	A1	$\therefore t_{max} = 0.35$ s
				Total 7 marks

Q2.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Integrates 0.138 t^2 twice	AO3.4	M1	$v = \int 0.138t^2 dt$ $= 0.046t^3 + c$ $t = 0, v = 0 \Rightarrow c = 0$ $s = \int 0.046t^3 dt$ $= 0.0115t^4 + k$
	Finds the correct expression for displacement condone no consideration of c	AO1.1b	A1	

	Demonstrates at least one constant of integration is zero	AO1.1b	B1	$t = 0$, $s = 0 \Rightarrow k = 0$ $0.0115t^4 = 100$ $t = 9.657$
	Finds the correct time for minibus A	AO1.1b	A1	
(b)	Integrates $0.024\ t^3$ twice	AO1.1a	M1	$v = \int 0.024t^3 \mathrm{d}t$ $= 0.006t^4 + c$ $t = 0$, $v = 0 \Rightarrow c = 0$ $s = \int 0.006t^4 \mathrm{d}t$ $= 0.0012t^5 + k$ $t = 0$, $s = 0 \Rightarrow k = 0$ $0.0012t^5 = 100$ $t = 9.642$ $9.642 < 9.657$ company chooses minibus B
	Finds the correct expression for displacement condone no consideration of c	AO1.1b	A1	
	Finds correct time for minibus B	AO1.1b	A1	
	States correct choice consistent with 'their' answers Must have integrated twice in both parts	AO3.2a	E1F	
(c)	Explains how reaction times of each driver could change the outcome	AO3.5b	E1	If Driver B's reaction time is greater than Driver A's then A could travel 100 metres faster than B
				Total 9 marks

Q3.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Differentiates, with at least one term correct	AO1.1a	M1	$\frac{\mathrm{d}v}{\mathrm{d}t} = 12t - 36^2$
	Selects and applies $F = ma$ to 'their' derivative Condone use of 400 for mass	AO1.1a	M1	$F = ma = 0.4\ (12t - 36t^2)$
	Obtains correct expression for force FT from 'their' $F = ma$ equation, provided the first M1 has been awarded (may be in factorised form)	AO1.1b	A1F	$= 4.8t - 14.4t^2$
(b)	Integrates v to find r, with	AO3.1b	M1	$r = \int \left(6t^2 - 12t^3\right) \mathrm{d}t$

at least one term correct			
Obtains correct integral (condone absence of c)	AO1.1b	A1	$r = 2t^3 - 3t^4 + c$
Deduces the value of c using initial conditions FT use of 'their' integral provided M1 awarded	AO2.2a	A1F	When $t = 0$, $r = 0$ so $0 = 2 \times 0^2 - 3 \times 0^4 + c$ so $c = 0$
Forms and solves an equation for t (condone numerical slip)	AO1.1a	M1	$0 = 2t^3 - 3t^4$ $= t^3(2 - 3t)$ $t = 0$ or $t = \frac{2}{3}$
Interprets solution, realising that the non-zero time is required (Must include units) FT use of 'their' equation for t provided both M1 marks have been awarded	AO3.2a	A1F	Next at O at $\frac{2}{3}$ seconds
			Total 8 marks

Q4.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Integrates both components with at least one correct	AO1.1a	M1	$\mathbf{r} = \int 40e^{-0.2t}\,dt\,\mathbf{i} + \int 50(e^{-0.2t} - 1)\,dt\,\mathbf{j}$ $= (-200e^{-0.2t} + c)\mathbf{i} + (-250e^{-0.2t} - 50t + d)\mathbf{j}$
	Obtains correct terms. (condone missing constants)	AO1.1b	A1	$t = 0$, $\mathbf{r} = 0\mathbf{i} + 0\mathbf{j} \Rightarrow c = 200$, $d = 250$ **OR**
	Evaluates both constants (or uses definite integration) using 'their' expression for **r**	AO3.4	M1	$\mathbf{r} = \int_0^t 40e^{-0.2t}\,dt\,\mathbf{i} + \int_0^t 50(e^{-0.2t} - 1)\,dt\,\mathbf{j}$ $= \left[-200e^{-0.2t}\mathbf{i} + (-250e^{-0.2t} - 50t)\mathbf{j}\right]_0^t$
	Obtains correct expression	AO1.1b	A1	$\mathbf{r} = 200(1 - e^{-0.2t})\mathbf{i}$ $+ (250 - 250e^{-0.2t} - 50t)\mathbf{j}$
(b)	Forms equation to find t based on horizontal component	AO3.4	M1	$200(1 - e^{-0.2t}) = 100$ $t = 5\ln 2 = 3.4657$ $y = 250 - 250e^{-0.2 \times 5\ln 2} - 50 \times 5\ln 2$ $= -48.3$
	Obtains correct time	AO1.1b	A1	

	Substitutes 'their' time into vertical component	AO1.1a	M1	
	Obtains correct displacement for 'their' time (to 1 sf, must have metres) FT only if both M1 marks have been awarded	AO3.2a	A1F	The parachutist has a vertical displacement of 50 m below the origin
(c)	Identifies vertical component of the velocity as first step	AO2.4	M1	$\frac{d}{dt}(50(e^{-0.2t}-1)) = -10e^{-0.2t}$ As there is no initial vertical component of velocity (and hence no air resistance) the initial acceleration is only due to gravity. Hence g is taken as 10 m s^{-2}
	Differentiates vertical component of velocity correctly	AO1.1b	A1	
	Considers implication of initial motion and reaches correct conclusion	AO2.2a	R1	
				Total 11 marks

Q5.

(a) $\mathbf{a} = \frac{d\mathbf{v}}{dt}$

M1 for either term correct

M1

$= 4\pi\sin\left(\frac{\pi}{3}t\right)\mathbf{i} - 18t\,\mathbf{j}$

Accept $12 \times \frac{\pi}{3}\sin\left(\frac{\pi}{3}t\right)\mathbf{i} - 18t\,\mathbf{j}$

Condone no ***i*** *in (a)*

A1

2

(b) (i) Using $\mathbf{F} = m\mathbf{a}$:

$\mathbf{F} = 4 \times \left[-4\pi\sin\left(\frac{\pi}{3}t\right)\mathbf{i} - 18t\,\mathbf{j}\right]$

Or either term correct

M1

$\mathbf{F} = -16\pi\sin\left(\frac{\pi}{3}t\right)\mathbf{i} - 72t\,\mathbf{j}$

A1

2

(ii) When $t = 3$, $\mathbf{F} = 4 \times [-4\pi \sin(\pi)\,\mathbf{i} - 54\mathbf{j}]$

$= -216\mathbf{j}$

B1

Magnitude is 216

ft finding magnitude of their F

B1ft

2

(c) $\mathbf{r} = \int \mathbf{v}\,\mathrm{d}t$

either term correct

M1

$= \frac{36}{\pi}\sin\left(\frac{\pi}{3}t\right)\mathbf{i} - 3t^3\,\mathbf{j} + \mathbf{c}$

No need for $\mathbf{c}$ *(otherwise CAO)*

Condone $\frac{12}{\left(\pi/3\right)}$

A1

When $t = 3$, $\mathbf{r} = 4\mathbf{i} - 2\mathbf{j}$

M1

$\rightarrow -81\mathbf{j} + \mathbf{c} = 4\mathbf{i} - 2\mathbf{j}$

$\mathbf{c} = 4\mathbf{i} + 79\mathbf{j}$

A1

$\mathbf{r} = \left\{\frac{36}{\pi}\sin\left(\frac{\pi}{3}t\right) + 4\right\}\mathbf{i} + \{79 - 3t^3\}\mathbf{j}$

CAO

A1

5

[11]

Q6.

(a) $v = \frac{\mathrm{d}s}{\mathrm{d}t}$

M1

$= 24t^2$

A1

2

(b) $a = \frac{dv}{dt}$

$= 48t$

B1

When $t = 2$, $a = 96$

B1

Using $F = ma$

$F = 3 \times 96$

M1

$= 288$ N

A1

4

[6]

Q7.

(a) $v = \int a \, dt$

$= (20t^2 + t^3)\mathbf{i} - 5e^{-4t}\,\mathbf{j} + \mathbf{c}$

M1 for either term correct.
*Condone no '+ **c**'.*

M1A1

When $t = 1$,

$6\mathbf{i} - 5e^{-4}\mathbf{j} = 21\mathbf{i} - 5e^{-4}\mathbf{j} + \mathbf{c}$

*Finding '+ **c**'; not using **c** = 6**i** – 5e*$^{-4}$ ***j**.*

M1

$\mathbf{c} = -15\mathbf{i}$

A1

$\mathbf{v} = (20t^2 + t^3 - 15)\mathbf{i}\ 5e^{-4t}\,\mathbf{j}$

A1

5

(b) When $t = 0$, $\mathbf{v} = -15\mathbf{i} - 5\mathbf{j}$

M1

Speed is $\sqrt{15^2 + 5^2}$

M1

$= 15.8 \text{ m s}^{-1}$

Accept $5\sqrt{10}$

A1

3

[8]

Q8.

(a) Using $F = ma$

$1600 \dfrac{dv}{dt} = 4000 - 40v$

$\dfrac{dv}{dt} = \dfrac{4000 - 40v}{1600}$

M1

$\dfrac{dv}{dt} = \dfrac{100 - v}{40}$

A1

2

(b) $40 \dfrac{dv}{100 - v} = dt$

B1

$40 \int \dfrac{dv}{100 - v} = \int dt$

M1

$-40 \ln(100 - v) = t + c$

Condone lack of '+ c'

A1

When $t = 0$, $v = 0 \Rightarrow c = -40 \ln 100$

$-40 \ln(100 - v) = t - 40 \ln 100$

$t = 40 \ln \dfrac{100}{100-v}$

$e^{\frac{t}{40}} = \dfrac{100}{100-v}$

M1A1

$v = 100 - 100\, e^{-\frac{t}{40}}$ or $100(1 - e^{-\frac{t}{40}})$

A1

6

[8]

Q9.

(a) using $\mathbf{F} = m\mathbf{a}$:

ie dividing by 50

M1

$\mathbf{a} = (6t - 1.2\,t^2)\,\mathbf{i} + 2\,e^{-2t}\,\mathbf{j}$

A1

2

(b) $\mathbf{v} = \int \mathbf{a}\,dt$

$= (3\,t^2 - 0.4\,t^3)\,\mathbf{i} - e^{-2t}\mathbf{j} + \mathbf{c}$

when $t = 0$, $\mathbf{r} = 7\,\mathbf{i} - 4\,\mathbf{j}$

condone lack of + c; M1 one term correct

M1A1

$\mathbf{c} = 7\,\mathbf{i} - 3\mathbf{j}$

$\mathbf{v} = (7 + 3\,t^2 - 0.4\,t^3)\,\mathbf{i} - (3 + e^{-2t})\,\mathbf{j}$

*ft from ke^{-2t} in (b); just adding 7**i** – 4**j**, m0 accept unsimplified. CAO*

m1A1

4

(c) when $t = 1$, $\mathbf{v} = 9.6\,\mathbf{i} - 3.135\,\mathbf{j}$

ft from (b)

M1A1

speed $= \sqrt{9.6^2 + 3.135^2}$

m1

$= 10.1\ \text{ms}^{-1}$

ft from (b)

A1

4

[10]

Q10.

(a) using $F = ma$

$0.4\dfrac{dv}{dt} = 2 - 4v$

M1

$\dfrac{dv}{dt} = -10(v - 0.5)$

Needs line above

A1

2

(b) hence $\int \dfrac{1}{v - 0.5}\,dv = -\int 10\,dt$

$\ln(v - 0.5) = -10t + c$

M1 for any side integrated correctly

M1A1

m1 for + c (and M1 gained)

m1

$v - 0.5 = Ce^{-10t}$

$t = 0,\ v = 1$

$\therefore C = 0.5$

A1

$\therefore v = 0.5 + 0.5e^{-10t}$

condone $v = 0.5 + e^{-10t-0.693}$

A1

5

(c) when $v = 0.55$, $0.55 = 0.5 + 0.5e^{-10t}$

substitute 0.55 into C's (b), after finding c, possible numerical error

M1

$10 = e^{10t}$
$t = \ln 10 \div 10$

A1

$= 0.230$

A1

3

[10]

Q11.

(a) (i) $a = \frac{dv}{dt}$

$= 12t + 8e^{-4t}$ m s^{-2}

M1 for either term correct

M1A1

2

(ii) When $t = 0.5$, $a = 6 + 8 \times e^{-2}$

m1

$= 7.08$ m s^{-2}

Condone 7.07

SC1 for 7.1 with no working

A1

2

(b) Using $F = ma$:
$F = 4 \times 7.08$
$= 28.3$ N

Ft from value awarded A1

B1ft

1

(c) $r = \int v\,dt$

At least two terms correct

M1

$= 2t^3 + \frac{1}{2}e^{-4t} + 8t + c$

Does not need + c

A1

When $t = 0$, $r = 0 \rightarrow c = -\frac{1}{2}$

Does not need $c = -\frac{1}{2}$

m1

$r = 2t^3 + \frac{1}{2}e^{-4t} + 8t - \frac{1}{2}$

Need r, s (or words)

A1

4

[9]

Q12.

(a) Using $F = ma$:

$m\frac{dv}{dt} = 49 - 9.8v$ or $5g$ 9.8 □v

Need to see $m\frac{dv}{dt}$ *or* $5\frac{dv}{dt}$ *or* $a = \frac{49-9.81}{5}$

M1

$\therefore \frac{dv}{dt} = -1.96(v - 5)$

Must see m terms (not a =…)

A1

2

(b) $\int \frac{dv}{v-5} = -1.96 \int dt$

And one side integrated

M1

$\ln(v - 5) = -1.96t + c$

Need + c, A1 each side

A1A1

When $t = 0$, $v = 7 \Rightarrow c = \ln 2$

OE

A1

$\ln \frac{v-5}{2} = -1.96t$

$\frac{v-5}{2} = e^{-1.96t}$

$v = 5 + 2e^{-1.96t}$

CAO

A1

5

[7]

Q13.

(a) $\mathbf{r} = \int v \, dt = (4t + t^3)\mathbf{i} + (12t - 4t^2)\mathbf{j} + \mathbf{c}$

*M1: either **i** or **j** term correct.*
*Condone no **c***

M1A1

When $t = 0$, $\mathbf{r} = 5\mathbf{i} - 7\mathbf{j}$
$\mathbf{c} = 5\mathbf{i} - 7\mathbf{j}$

*Any attempt at **c***

M1

$\mathbf{r} = (5 + 4t + t^3)\mathbf{i} + (-7 + 12t - 4t^2)\mathbf{j}$

A1

4

(b) $\mathbf{a} = \frac{dv}{dt}$

$\mathbf{a} = 6t\,\mathbf{i} - 8\,\mathbf{j}$

M1: either term correct

M1A1

2

(c) Using $\mathbf{F} = m\mathbf{a}$

Or: using $\mathbf{F} = m\mathbf{a}$

M1

$\mathbf{F} = 2(6t\,\mathbf{i} - 8\,\mathbf{j})$

$\mathbf{F} = 2(6t\,\mathbf{i} - 8\,\mathbf{j})$

$= 12t\,\mathbf{i} - 16\,\mathbf{j}$

*When t = 1, **F** = 12 **i** – 16 **j***

A1

∴ Magnitude of force is $(144t^2 + 256)^{\frac{1}{2}}$ when t = 1

Magnitude of force is $(144 + 256)^{\frac{1}{2}}$

M1

= 20 N

= 20 N

A1

4

[10]

Q14.

(a) (i) $F = 65g - 260v$

Accept $260v - 65g$

$= 65(9.8 - 4v)$

AG must see $65v$ or 260

B1

1

(ii) Using $F = ma$

$65\frac{\mathrm{d}v}{\mathrm{d}t} = 65(9.8 - 4v)$

Need to see terms in m (condone – sign)

M1

$\frac{\mathrm{d}v}{\mathrm{d}t} = -4(v - 2.45)$

AG

A1

2

(b) $\frac{1}{v - 2.45}\frac{\mathrm{d}v}{\mathrm{d}t} = -4$

B1

$$\int \frac{1}{v-2.45}\,\mathrm{d}v = -\int 4\,\mathrm{d}t$$

$\ln(v - 2.45) = -4t + c$

M1: log side correct

M1

$-4t + c$

A1

$v - 2.45 = Ce^{-4t}$

$t = 0$, $v = 19.6$

$\therefore C = 17.15$ or $e^{2.84}$

Or $c = \ln 17.15$ or 2.84

A1

$\therefore v = 2.45 + 17.15e^{-4t}$ $2.45 + 17.2e^{-4t}$

A1

5

[8]

Q15.

(a) $\mathbf{a} = \frac{\mathrm{d}v}{\mathrm{d}t}$

$\mathbf{a} = -8e^{-2t}\,\mathbf{i} + (6 - 6t)\mathbf{j}$

*M1: Differentiating with either of the two components correct. Do not need to see **i** or **j**.*

*A1: Correct **i** component.*

*A1: Correct **j** component.*

M1
A1
A1

3

(b) (i) Using $\mathbf{F} = m\mathbf{a}$

$\mathbf{F} = 5 \times \{-8e^{-2t}\,\mathbf{i} + (6 - 6t)\mathbf{j}\}$

$= -40e^{-2t}\,\mathbf{i} + (30 - 30t)\mathbf{j}$

M1: Multiplying their acceleration by 5, even if not a vector.

A1: Correct expression.

M1
A1
2

(ii) Magnitude of **F** is

$\{(-40)^2 + (30)^2\}^{\frac{1}{2}}$

M1: Finding magnitude from two non-zero terms. Must add terms and square root. Condone $\{(40)^2 + (30)^2\}^{\frac{1}{2}}$
M1

= 50

A1: Correct answer only.
In this part, condone lack of negative signs in expression for force in (b) (i).
A1
2

(c) When **F** acts due west, **j** component is zero

$30 - 30t = 0$

M1: Putting $\mathbf{j}$ *component equal to zero.*
M1

$t = 1$

A1: Correct time.
A1
2

(d) $\mathbf{r} = -2e^{-2t}\,\mathbf{i} + (3t^2 - t^3)\mathbf{j} + \mathbf{c}$

M1: Integration with either of the two components correct. Do not need to see $\mathbf{i}$ *or* $\mathbf{j}$*.*
A1: Correct $\mathbf{i}$ *component.*
A1: Correct $\mathbf{j}$ *component.*
Condone lack of $+\,\mathbf{c}$
M1
A1
A1

When $t = 0$, $\mathbf{r} = 6\mathbf{i} + 5\mathbf{j}$ $\therefore \mathbf{c} = 8\mathbf{i} + 5\mathbf{j}$

dM1: Finding $\mathbf{c}$ *using* $6\mathbf{i} + 5\mathbf{j}$ *and* $e^0 = 1$*.*
dM1

$\therefore \mathbf{r} = (8 - 2e^{-2t})\mathbf{i} + (5 + \frac{1}{4}t^4 - t^3)\mathbf{j}$

A1: Correct position vector.
A1
5

[14]

Q16.

$v = \frac{dt}{dv}$

M1 for either $\frac{ds}{dt}$ *or 1 of 2 terms correct (ignore signs)*

M1

$= 10t - 12 \sin 4t$

A1A1

[3]

Q17.

(a) Using $\mathbf{F} = m\mathbf{a}$,

$400 \cos \frac{\pi}{2} t\, \mathbf{i} + 600t^2\, \mathbf{j} = 200\, \mathbf{a}$

M1

$\mathbf{a} = 2 \cos \frac{\pi}{2} t\, \mathbf{i} + 3t^2\, \mathbf{j}$

A1

2

(b) $\mathbf{v} = \int a\, dt$

M1 for either $\int a\, dt$ *or 1 of 2 terms correct*

M1

$= \frac{4}{\pi} \sin \frac{\pi}{2} t\, \mathbf{i} + t^3 \mathbf{j} + \mathbf{c}$

m1 for + ***c***

A1m1

When $t = 4$, $\mathbf{r} = -3\mathbf{i} + 56\mathbf{j}$,
$64\mathbf{j} + \mathbf{c} = -3\mathbf{i} + 56\mathbf{j}$

m1

$\therefore\ \mathbf{c} = -3\,\mathbf{i} - 8\,\mathbf{j}$

$\therefore\ \mathbf{v} = (\frac{4}{\pi} \sin \frac{\pi}{2} t - 3)\,\mathbf{i} + (t^3 - 8)\,\mathbf{j}$

Do not accept $\frac{\frac{2}{\pi}}{2}$ *Accept 1.27 for* $\frac{4}{\pi}$

A1

5

(c) When particle is moving due west, northerly component is zero

M1

$\therefore t^3 - 8 = 0$

A1ft

$t = 2$

A1

3

(d) When $t = 2$, $\mathbf{v} = -3\mathbf{i} + 0\mathbf{j}$

B1ft

Speed of particle is 3 m s^{-1}

B1 for change –3 to +3

B1

2

[12]

Q18.

$$\frac{\mathrm{d}v}{\mathrm{d}t} = -\frac{\lambda}{v^{\frac{1}{4}}}$$

M1

$$\int v^{\frac{1}{4}}\,\mathrm{d}v = -\int \lambda\,\mathrm{d}t$$

Condone one of $v^{-\frac{1}{4}}, +\int \lambda \mathrm{d}t, \frac{1}{\lambda}$

m1

$$\frac{4}{5}v^{\frac{5}{4}} = -\lambda t + c$$

m1 for + c

A1A1m1

$t = 0, v = u \therefore c = \frac{4}{5}u^{\frac{5}{4}}$

A1

$\therefore v^{\frac{5}{4}} = u^{\frac{5}{4}} - \frac{5}{4}\lambda t$

$v = \left(u^{\frac{5}{4}} - \frac{5}{4}\lambda t\right)^{\frac{4}{5}}$

A1

[7]

Q19.

$r = \int v \ \mathrm{d}t$

M1

$= t^4 + 4\cos 2t + 5t\ (+\ c)$

A1

When $t = 0$, $r = 0 \Rightarrow c = -4$

Finding c correctly

M1

$\therefore \ r = t^4 + 4\cos 2t + 5t - 4$

A1ft

4

[4]

Q20.

(a) $\mathbf{v} = \frac{\mathrm{d}r}{\mathrm{d}t}$

M1

$\mathbf{v} = (\mathrm{e}^{\frac{1}{2}t} - 8)\mathbf{i} + (2t - 6)\mathbf{j}$

***i** terms*

A1

j terms

A1

3

(b) (i) When t = 3, $\mathbf{v} = -3.52\mathbf{i}$

Accept $(e^{\frac{3}{2}} - 8)\mathbf{i}$

B1

Speed is 3.52 m s^{-1}

3.5 does not give 2nd B mark

B1

2

(ii) West

B1

1

(c) $\mathbf{a} = \frac{1}{2}e^{\frac{1}{2}t}\mathbf{i} + 2\mathbf{j}$

M1A1

When t = 3, $\mathbf{a} = \frac{1}{2}e^{\frac{3}{2}}\mathbf{i} + 2\mathbf{j}$ or $2.24\mathbf{i} + 2\mathbf{j}$

A1

3

(d) Using $\mathbf{F} = m\mathbf{a}$:

Accept $\boldsymbol{F} = 7\boldsymbol{a}$

M1

$\mathbf{F} = 7\left(\frac{1}{2}e^{\frac{3}{2}}\mathbf{i} + 2\mathbf{j}\right)$

$\therefore$ Magnitude of force is $7\left((\frac{1}{2}e^{\frac{3}{2}})^2 + 2^2\right)^{\frac{1}{2}}$

M1

$\mathbf{F} = 21.025$

$\mathbf{F} = 21.0$

Accept 21

A1

3

[12]

Q21.

(a) $\mathbf{a} = \frac{d\mathbf{v}}{dt} = (3t^2 - 15)\mathbf{i} + (6 - 2t)\mathbf{j}$

*A1 (**i** terms) A1 (**j** terms)*

M1A1A1

3

(b) (i) Using $\mathbf{F} = m\mathbf{a}$:
Force = $4 \times \{(3t^2 - 15)\mathbf{i} + (6 - 2t)\mathbf{j}\}$

M1

$= (12t^2 - 60)\mathbf{i} + (24 - 8t)\mathbf{j}$

AG

A1

2

(ii) When $t = 2$, force $= -12\mathbf{i} + 8\mathbf{j}$

M1A1

Magnitude of force = $\sqrt{12^2 + 8^2}$ N

M1

= 14.4 (N)

A1

4

[9]

Q22.

(a) (i) $a = \frac{dv}{dt} = 6t - 6\cos 3t$

M1 for at least one term correct

M1A1

2

(ii) When $t = \frac{\pi}{3}, a = 6 \times \frac{\pi}{3} - 6\cos(3.\frac{\pi}{3})$

M1

$= 2\pi + 6$

AG

A1

2

(b) $r = t^3 + \frac{2}{3}\cos 3t + 6t + c$

M1 for 3 terms including cos3t term
Condone no '+ c'

M1A1

When $t = 0$, $r = 0$ $\therefore c = -\frac{2}{3}$

M1

$\therefore r = t^3 + \frac{2}{3}\cos 3t + 6t - \frac{2}{3}$

A1

4

[8]

Q23.

(a) $\mathbf{v} = \frac{\mathrm{d}r}{\mathrm{d}t}$

$\mathbf{v} = (3t^2 - 6t)\mathbf{i} + (4 + 2t)\mathbf{j}$

M1A1

2

(b) (i) $\mathbf{a} = (6t - 6)\mathbf{i} + 2\mathbf{j}$

M1A1ft

Using $\mathbf{F} = m\mathbf{a}$:
$\mathbf{F} = (18t - 18)\mathbf{i} + 6\mathbf{j}$

A1ft

3

(ii) When $t = 3$, $\mathbf{F} = 36\mathbf{i} + 6\mathbf{j}$

Magnitude is $\sqrt{36^2 + 6^2}$

M1

= 36.5

Accept $6\sqrt{37}$ *; ft from (b)(i)*

A1ft

2

(c) When **F** acts due north:
Component of **F** in the **i** direction is 0

M1

$18t - 18 = 0$
$t = 1$

ft from (b)(i)

A1ft

2

[9]

Q24.

(a) $a = \frac{\mathrm{d}v}{\mathrm{d}t} = 12t + 4$

M1 A1

2

(b) Using $\mathbf{F} = m\mathbf{a}$,
Force = $3 \times (12t + 4)$

M1

When $t = 4$, force = $3\,(12 \times 4 + 4)$
Force = 156 N

A1

2

[4]

Q25.

(a) $a = \frac{\mathrm{d}v}{\mathrm{d}t} = 12t + 4$

M1A1

2

(b) Using $F = ma$,
Force = 3 × (12t + 4)

M1

When t = 4, force = 3 (12 × 4 + 4)
Force = 156 N

A1

2

(c) $r = 2t^3 + 2t^2 - 7t + c$

M1A1

When t = 0, r = 5, $\therefore$ c = 5

M1

$\therefore$ $r = 2t^3 + 2t^2 - 7t + 5$

SC3 if no '+c' seen

A1

4

[8]

Q26.

(a) $\mathbf{v} = 1.2\mathbf{i} + (-0.9 \sin t)\mathbf{j}$

Convincingly differentiation

M1A1

2

(b) (i) Speed = $\sqrt{1.2^2 + 0.9^2 \sin^2 t}$

M1A1

2

(ii) Maximum speed = $\sqrt{1.2^2 + 0.9^2}$

Use of sint = 1

m1

= 1.5

A1

2

[6]

Q27.

(a) Using $F = ma$:
$2400\mathbf{i} - 4800t\mathbf{j} = 800\mathbf{a}$

M1

$\mathbf{a} = 3\mathbf{i} - 6t\mathbf{j}$

A1

2

(b) $\mathbf{v} = \int \mathbf{a}\ \mathrm{d}t$

M1

$= 3t\mathbf{i} - 3t^2\mathbf{j} + \mathbf{c}$
*Condone no '+ **c**'*

A1

When $t = 0$, $\mathbf{v} = 6\mathbf{i} + 30\mathbf{j}$
$\therefore\ \mathbf{c} = 6\mathbf{i} + 30\mathbf{j}$
*Needs '+ **c**' above*

M1

$\therefore\ \mathbf{v} = (3t + 6)\mathbf{i} + (30 - 3t^2)\mathbf{j}$
AG

A1

4

(c) $\mathbf{r} = \int \mathbf{v}\ \mathrm{d}t$

M1

$= (\frac{3}{2}t^2 + 6t)\,\mathbf{i} + (30t - t^3)\,\mathbf{j} + \mathbf{d}$
*A1 **i** term, A1 **j** term; condone no '+ **d**'*

A1,A1

When $t = 0$, $\mathbf{r} = 2\mathbf{i} + 5\mathbf{j}$
$\therefore\ \mathbf{d} = 2\mathbf{i} + 5\mathbf{j}$

M1

$\therefore\ \mathbf{r} = (\frac{3}{2}t^2 + 6t + 2)\mathbf{i} + (30t - t^3 + 5)\mathbf{j}$

A1

5

[11]

Q28.

(a) (i) $a = 2 + 12e^{-t}$

Differentiating, with at least one term correct.
Correct velocity

M1A1

2

(ii) $2 < a \leq 14$

For 2, For 14

B1, B1

Correct inequalities

B1

3

(b) $s = t^2 + 12e^{-t} + c$

Integrating, with at least one term correct.

M1

Correct expression with or without c

A1

$s = 0, t = 0 \Rightarrow c = -12$

Finding c

dM1

$s = t^2 + 12e^{-t} - 12$

Correct final expression

A1

4

[9]

Q29.

(a) $\mathbf{F} = 12\cos t\mathbf{i} - 30\sin t\mathbf{j}$

Use of $\mathbf{F} = m\mathbf{a}$

M1

$\mathbf{F}(0) = 6 \times 2\mathbf{i}$ so $F = 12$ N

Correct $\mathbf{F}$

A1

Correct magnitude

A1

3

(b) $\mathbf{v} = \int 2\cos t\,dt\,\mathbf{i} + \int -5\sin t\,dt\,\mathbf{j}$

Integrating

M1

$= (2\sin t + c_1)\mathbf{i} + (5\cos t + c_2)\mathbf{j}$

Correct **i** *component*

A1

Correct **j** *component*

A1

Both of above with or without constants

$\mathbf{v}(0) = 2\mathbf{i} + 10\,\mathbf{j} \Rightarrow c_1 = 2,\ c_2 = 5$

Finding constants of integration

dM1

$\mathbf{v} = (2\sin t + 2)\mathbf{i} + (5\cos t + 5)\mathbf{j}$

Correct final answer

A1

5

[8]

Q30.

(a) $\mathbf{v} = (6t^2 - 2t)\mathbf{i} + (1 - 12t^2)\mathbf{j}$

differentiating both components

M1

one component correct

A1

second component correct

A1

3

(b) (i) $\mathbf{v}\left(\frac{1}{3}\right) = \left(\frac{6}{9} - \frac{2}{3}\right)\mathbf{i} + \left(1 - \frac{12}{9}\right)\mathbf{j} = -\frac{1}{3}\mathbf{j}$

substituting the value for t into their ***v***

M1

correct velocity

A1

2

(ii) Travelling due south

correct description (Follow through from $\mathbf{v} = \pm k\mathbf{j}$ *)*

A1ft

1

(c) $\mathbf{a} = (12t - 2)\mathbf{i} - 24t\mathbf{j}$

differentiating their velocity

M1

$\mathbf{a}(4) = 46\mathbf{i} - 96\mathbf{i}$

correct acceleration at time t

A1

correct acceleration at t = 4

A1

3

(d) $\mathbf{F} = 6(46\mathbf{i} - 96\mathbf{j}) = 276\mathbf{i} - 576\mathbf{j}$

apply Newton's second law correctly

M1

$F = \sqrt{276^2 + 576^2} = 639 \text{ N}$

finding magnitude

M1

correct magnitude

A1

3

or

$a = \sqrt{46^2 + 96^2} = 106.45$

$F = 6 \times 106.45 = 639 \text{ N}$

[12]

Q31.

(a) $F = 800 + \frac{1200}{20}t = 800 + 60t$

finding the gradient of the line

M1

correct gradient

A1

$1200a = 800 + 60t$

correct intercept

B1

$a = \frac{800}{1200} + \frac{60}{1200}t = \frac{2}{3} + \frac{t}{20}$

using Newton's second law on two terms

dM1

AG

correct result from correct working

A1

5

(b) $v = \int \frac{2}{3} + \frac{t}{20}\,dt = \frac{2t}{3} + \frac{t^2}{40} + c$

integrating

M1

correct integral with or without c

A1

$v = 0, t = 0 \Rightarrow c = 0$

A1

$v = \frac{2t}{3} + \frac{t^2}{40}$

showing c = 0

A1

3

(c) $s = \int_0^{20} \frac{2t}{3} + \frac{t^2}{40}\,dt$

integrating

M1

correct integral, with or without c.

A1

$= \left[\frac{t}{3} + \frac{t^3}{120}\right]_0^{20}$

use of both limits or finding c

dM1

= 200 m

correct distance

A1

4

(d) The $\frac{2t}{3}$ term would change, because only

correct term

B1

the constant term in the force would change.
When integrated this becomes the t term in the velocity.

correct explanation

B1

2

[14]

Q32.

(a) **a** = 8 cos t **i** – 4 sin t **j**

M1 differentiation

M1A1

F = 2 cos t **i** – sin t **j**

A1

3

(b) (i) | F | = $\sqrt{(4\cos^2 t + \sin^2 t)}$

Magnitude

M1

$= \sqrt{(3\cos^2 t + 1}$

m1

CAO

A1

3

(ii) $1 \le |F| \le 4$

B1 each value (at end of range)

B1B1

2

[8]

Q33.

(a) (i) $\mathbf{v} = 2\cos 2t\mathbf{i} + 6\mathbf{j}$

M1 differentiation (6t)

M1A1

2

(ii) $|\mathbf{v}| = \sqrt{4\cos^2 2t + 36}$

Sum of squares, for v or v^2

M1

ft trig term ***v***

A1F

CAO

A1

3

(iii) $\cos^2 2t = 0$ or $\cos 2t = 0$

M1

$t = \frac{\pi}{4}$

radians

A1

2

(b) (i) $\boldsymbol{a} = -4\sin 2t\,\mathbf{i}$

Differentiation attempt

M1

$\boldsymbol{F} = 0.25\boldsymbol{a}$

Used

M1

$\boldsymbol{F} = -\sin 2t\,\mathbf{i}$

ft ***v****, see vector*

A1F

3

(ii) Direction is $\pm$ **i**

B1

$|\sin 2t| \le 1$

B1

2

[12]

Q34.

Answers in (a), (b) and (c)(i) must be vectors

(a) $\mathbf{a} = 4t\mathbf{i} + 3\mathbf{j}$

Must be vector, both i and j present

B1

1

(b) $\mathbf{v} = 2t^2\mathbf{i} + 3t\mathbf{j}$

Integration of acceleration attempted

M1

*ft **a**, both **i** and **j** present*

A1F

2

(c) (i) $\mathbf{r} = \int_0^t \left(2t^2\mathbf{i} + 3t\mathbf{j}\right)dt$

Integration attempted

M1

$= \dfrac{2t^3}{3}\mathbf{i} + \dfrac{3t^2}{2}\mathbf{j}$

*A1 each term. Condone **i** and **j** missing.*
–1 for constants present,
–1 for both terms unsimplified

A1A1

3

(ii) $\left.\begin{array}{l}\mathbf{v} \text{ parallel to } \mathbf{i} + \mathbf{j} \\ 2t^2 = 3t\end{array}\right\}$

M1

$t = \frac{3}{2}$

Solve, ft ***v*** *with vector components*

A1F

$\mathbf{r} = \frac{9}{4}\mathbf{i} + \frac{27}{8}\mathbf{j}$

A1F

Distance $= \sqrt{\left(\left(\frac{9}{4}\right)^2 + \left(\frac{27}{8}\right)^2\right)}$

Magnitude of ***r*** *in any form*

M1

= 4.06 (4.056)

Numerical answer, ft candidate's ***r*** *and* t

A1F 5

[11]

Q35.

(a) $\mathbf{v} = 6\mathbf{i} + 2t\,\mathbf{j}$

M1: differentiation attempted and vector quantity for ***v*** *given*

M1A1 2

(b) sp = $\sqrt{(6^2 + 4t^2)}$ ms^{-1}

M1: sum of squares attempted giving scalar expression

A1 : all correct, accept $(2t)^2$

f.t. ***v*** *with 2 components*

M1A1F 2

(c) $\sqrt{(6^2 + 4t^2)} = 6\sqrt{2}$

o.e.; scalar expression for ***v*** *in terms of t from (b) used*

M1

$36 + 4t^2 = 36 \times 2$

$t = 3$

f.t. minor slip in (b) provided t is positive solution of quadratic eqn

A1F

2

[6]